Massless and Massive 4D Spinor-Helicity

$\underline{p^2 = 0}$:

$$\text{Det}[p] = 0 \implies \text{Rank}[p] = 1 \implies p^{a\dot{a}} =: {}^{a}|p\rangle[\tilde{p}|^{\dot{a}}$$

$|p\rangle, |\tilde{p}]$ are U(1) covariant *Weyl* spinors

$\underline{p^2 \neq 0}$:

$$\text{Det}[p] \neq 0 \implies \text{Rank}[p] = 2 \implies p^{a\dot{a}} = {}^{a}|p\rangle[\tilde{p}|^{\dot{a}} + {}^{a}|q\rangle[\tilde{q}|^{\dot{a}}$$

Rearranging a bit,

$$\begin{aligned} p &= {}^{a}|p\rangle[\tilde{p}|^{\dot{a}} + {}^{a}|p][\tilde{p}|^{\dot{a}} \\ &= \begin{pmatrix} |p\rangle & |p] \end{pmatrix} \cdot \begin{pmatrix} [\tilde{p}| \\ \langle\tilde{p}| \end{pmatrix} =: {}^{a}|\mathbf{p}\rangle^{I}{}_{I}[\tilde{\mathbf{p}}|^{\dot{a}} \end{aligned}$$

$|\mathbf{p}\rangle, |\tilde{\mathbf{p}}]$ are SU(2) covariant *Dirac* spinors
Lots to say about the choice of basis, ask me later!



Covariant BCFW for all masses

We upgrade the usual mBCFW shift

$$|\hat{a}\rangle = |a\rangle + z|b\rangle, \qquad |\hat{\tilde{b}}] = |\tilde{b}] - z|\tilde{a}]$$

$$\Longrightarrow {}^{I}|\hat{\mathbf{a}}\rangle = {}^{I}|\mathbf{a}\rangle + z\Xi^{IJ}{}_{J}|\mathbf{b}\rangle, \qquad {}^{J}|\hat{\tilde{\mathbf{b}}}] = {}^{J}|\tilde{\mathbf{b}}] - z\Xi^{IJ}{}_{I}|\tilde{\mathbf{a}}]$$

Since Ξ^{IJ} is null it suffices to give (with $p^\flat := \lim_{m^2\to 0} p$)

$$\Xi^{11} = 1, \quad \Xi^{12} = \frac{{}^{K}\langle \mathbf{a}b\rangle\ {}_{K}[\tilde{\mathbf{a}}\tilde{b}\rangle}{|a\wedge b| + \frac{1}{2}(a\cdot b^\flat)}, \quad \Xi^{21} = \frac{[a\mathbf{b}\rangle^{K}\ [\tilde{a}\tilde{\mathbf{b}}]_{K}}{|a\wedge b| + \frac{1}{2}(a^\flat\cdot b)}$$

This formula is smooth in all massless limits for a and/or b, yielding the proper BCFW shift for *any* pair of masses

A massive shift is given in 2112.12312 but requires using a special basis, in which our expressions match. The above formula is the first *covariant* massive shift.



Ok, so what?

Applications and related questions:

- Analytic MBCFW (code in-development)
  (coulomb branch, mass regulators, classical scattering, ...)
  (massive amplitudes with *rational* kinematics)
- Geometric understanding of MBCFW
  ($|a \wedge b|$, ...)
- Massless limits and basis choices of MSH expressions
  (NOT always as simple as "unbold the spinors")
  (vDVZ discontinuity in massive SUGRA, ...)

Quarendo invenietes - by seeking you will discover
Details galore in publications to come, stay tuned!

